

PVTRACE · CALCULATION NOTE

PVTrace sizing calculations

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
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REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	R3 row notes the capacitor selection rule and section 7 the accepted mass margin, decided on 2026-10-02 (PVT-DEC-001, items 3 and 4); no number changed
0.3	2026-10-01	Amish Chadha	Constructable design (PVT-DDR-003); mass and cost updated; R17 reported against the value-engineering target
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First TRL 3 sizing note (sweep model, range, resolution, error budget, STC uncertainty, discharge, heat, battery, mass, cost) for the 6.6 mF isolated design in PVT-DDR-001

PVTrace sizing calculations

On paper, PVTrace meets nine of its seventeen requirements (R1, R2, R4, R5, R10, R11, R12, R14, R15); R17 (cost) is reported against the value-engineering target and is USD 13 over it. Three are **not met**: R3 (sweep time) at worst-case capacitor tolerance, R13 (display readability in sun, not shown even with the sun hood, with heat at risk) and R16 (insulation screening, out of scope). R7 and R8 are **at risk** because they rest on the reference cell calibration, and R6 and R9 cannot be verified until hardware exists.

Three TRL 2 figures change. With the third capacitor (6.6 mF, PVT-DDR-001 decision 3) the high-current module sweeps in 20.5 ms at nominal capacitance but only 16.4 ms at the -20 % tolerance of electrolytic capacitors, so R3 is still not met in the worst case. The stored energy at 100 V rises from 22 J to 33 J. Battery life falls from about 16 h to 10.3 h once the isolation barrier (decision 4) and a 5 V boost are counted; R14 is still met. The module current at 99 % of Voc is up to 2.5 A, so the load switch must stay closed a few milliseconds past the end of the sweep to open below 0.5 A (R11).

Version 0.2 applies Amish's decisions of 2026-09-25 (PVT-DDR-002): the budget in `project.yaml` rises from \$150 to \$165, and a printed display sun hood (BOM line 15, +\$1, about 15 g) is added. The parts total goes from \$163 to \$164, so R17 changes from not met to met with a \$1 margin. Mass goes from 1.41 kg to 1.43 kg (R12 still met); the hood top is level with the isolator knob at 98 mm, so the envelope does not change. No other result changes.

Version 0.3 follows the constructable design of PVT-DDR-003: a 1.5 mm polycarbonate chassis plate, a USB-C charging socket, a smaller bonded window, a wider sun hood with screw tabs, a 16-bit ADC for the reference cell and added fixings. Mass goes from 1.43 kg to 1.49 kg (R12 still met, about 14 g to spare) and the estimated cost from USD 164 to USD 178. Following Amish's instruction of 2026-10-01, `budget_usd` (USD 165) is treated as a value-engineering target, not a limit, so R17 is reported as USD 13 over the target. The sweep, discharge, heat and battery results do not change.

Every number here is printed by `docs/04-calcs/sizing.py` (run from the repo root: `python docs/04-calcs/sizing.py`), which also writes `docs/04-calcs/results.csv`. The script reads the case dimensions from `cad/src/model.py` and the costs from `bom/bom.csv`. All values are first-principles estimates for a paper design; nothing is measured.

■ 1. Assumptions

Table 1. Inputs. All are assumptions for a paper design.

Input	Value	Basis
Module cases	36-cell 100 W (22.0 V, 6.0 A); 108 half-cell 410 W (37.5 V, 13.9 A; MPP 31.4 V, 13.06 A); 144 half-cell 450 W reference (49.5 V, 11.6 A; MPP 41.5 V, 10.85 A); 2 x 60-cell 250 W in series (75.2 V, 8.9 A); rating limit, the reference curve scaled to 100 V and 20 A	Datasheet-class values, PVT-REQ-001
Module model	Single-diode model, ideality 1.2 per cell, series and shunt resistance solved from Isc, Voc, MPP and zero dP/dV at MPP	Standard five-parameter fit
Load capacitance	3 x 2200 μ F 160 V = 6.6 mF, ± 20 %	Decision 3, PVT-DDR-001; typical electrolytic tolerance
Loop resistance	56.6 m Ω : leads 8.6, MC4 contacts 2.0, fuse 5.0, isolator 2.0, shunt 4.0, MOSFET 10.0, capacitor ESR 20.0, wiring 5.0 m Ω	Copper at 20 °C; typical part values
Sampling	25,000 V-I pairs per second, voltage and current converted one after the other (20 μ s apart)	Half the rate of a 100 kS/s two-channel ADC
Front end	1 M Ω and 39 k Ω divider (0.1 %, 25 ppm/°C), 4 m Ω shunt (50 ppm/°C), gain 50, 4.096 V reference (50 ppm/°C), 12-bit ADC with ± 1 LSB INL, 30 °C between calibration and use	Typical parts in bom/ bom.csv line 4
Voc temperature coefficient	-0.30 %/°C; coldest cell -10 °C	Typical crystalline silicon
STC terms	Pmax coefficient -0.35 %/°C; module temperature ± 2 °C; spectral and angle mismatch ± 2 %; reference cell ± 3 to ± 5 %	PVT-PRC-001; decision 6
Discharge	22 Ω active dump, 10 k Ω passive bleed; one sweep every 5 s at most	PVT-PRC-001
Power	Controller and lit TFT 0.60 W; isolated side 0.25 W; 5 V boost 90 %; 3,000 mAh 18650 at 3.6 V, 90 % usable; 20 s of controller time per sweep	TRL 2 estimate plus isolation
Heat	9 W/(m ² K) from the case outside; 1,000 W/m ² on the lid; solar absorptance 0.30 (light grey) or 0.90 (dark); 45 °C ambient	Still air, natural convection plus radiation

■ 2. Sweep (R3, R4)

Table 2 gives the sweep for each case. The sweep time is the time for the capacitor to rise from 0 V to 99 % of Voc, integrated as $t = \int C dV / I(V)$ along the fitted curve, with the loop resistance and the bleed and divider currents included. It scales in proportion to the load capacitance.

Table 2. Sweep with 6.6 mF (single-diode model)

Case	Sweep, nominal (ms)	Sweep, -20 % C (ms)	Sweep, 4.4 mF (ms)	V-I pairs, nominal	Energy stored (J)	First sample (V)	Samples per volt near Isc, -20 % C
36-cell 100 W	28.9	23.1	19.3	722	1.6	0.34	22.0
108 half-cell 410 W (high current)	20.5	16.4	13.7	513	4.5	0.79	9.5
144 half-cell 450 W (reference)	32.2	25.8	21.5	805	7.9	0.66	11.4
2 x 60-cell 250 W (high voltage)	67.5	54.0	45.0	1,687	18.3	0.50	14.8
Rating limit, 100 V, 20 A	37.7	30.1	25.1	942	32.3	1.13	6.6

- **R3 is not met at worst-case tolerance.** The shortest sweep is 20.5 ms at nominal capacitance but 16.4 ms if the three capacitors are 20 % low. Holding 20 ms at -20 % needs 8.0 mF rated (6.4 mF actual), which is a fourth capacitor. The longest sweep, the high-voltage case with +20 % capacitance, is about 81 ms, well inside 200 ms. The 4.4 mF column reproduces the TRL 2 estimates (13, 21 and 43 ms) within 2 ms, which checks the TRL 2 scaling.
- **R4 is met.** The fewest pairs are 410 (high-current case, -20 % capacitance), against 200.
- **Isc.** Loop resistance holds the module at 0.34 to 1.13 V at the first sample, 1 to 2 % of Voc, so firmware extrapolates Isc from the first samples. Pre-charging the bank negative is not needed.
- **Module capacitance.** With a capacitive load $dV/dt = I / C_{load}$, so the module's own capacitance C_{mod} causes a current error of about C_{mod} / C_{load} , independent of the sweep time. An error of 0.5 % or less needs C_{mod} of 26 μ F or less at the lowest load capacitance. The capacitance of TOPCon and HJT modules at MPP is not verified in this note; the 20 ms bound in R3 remains a working figure until a source or a measurement confirms it.
- **Sample skew.** Converting V and I 20 μ s apart gives up to 0.13 % error in Pmax at nominal capacitance (0.16 % at -20 %), in the high-current case. Firmware removes it by interpolating each voltage to the time of its current sample.

■ 3. Range, resolution and accuracy (R1, R2, R5, R6)

Range (R1, R2). Cold-weather Voc at a -10 °C cell temperature is 24.3 V, 41.4 V, 54.7 V and 83.1 V for the four cases, all below 100 V. Two 450 W reference modules in series reach 99.0 V at STC and 109.4 V at -10 °C, so the 100 V check refuses them; the manual must say so. At 100 V the capacitors have a 1.60 times voltage margin and the MOSFETs 1.50 times. At 20 A the shunt dissipates 1.6 W against its 3 W rating, and the load MOSFET conducts 0.15 J per sweep. Both are met by design review.

Resolution. The divider gives a voltage full scale of 109.1 V (26.6 mV per count); the shunt and amplifier give 20.48 A full scale (5.00 mA per count).

Accuracy (R5). After a two-point calibration, drift over 30 °C leaves ± 0.17 % of reading ± 0.024 % of full scale on voltage and ± 0.21 % of reading ± 0.026 % of full scale on current, inside ± 1 % ± 0.1 %. The amplifier offset before calibration is worth 6.25 mA, so the zero must be calibrated. R5 is met on paper; it depends on calibration against a meter and on the component grades in `bom/bom.csv` line 4.

Repeatability (R6). Quantization at the reference MPP is 0.079 % of Pmax per sample and the instrument contributes ± 0.27 % to Pmax, so the ± 1 % target will be set by irradiance drift between sweeps. R6 cannot be verified at TRL 3.

■ 4. Irradiance and STC uncertainty (R7, R8)

Table 3. STC Pmax uncertainty (root sum square)

Term	Value
Reference cell calibration	± 3 to ± 5 %
Spectral and angle mismatch	± 2 %
Module temperature, ± 2 °C at -0.35 %/°C	± 0.7 %
Instrument, V and I	± 0.27 %
Total	± 3.7 to ± 5.4 %

R8 (± 5 %) is met only if the reference cell calibration is within ± 4.5 %; R7 (± 5 % irradiance) needs ± 4.6 %. Both are **at risk** until the once-calibrated reference cell (decision 6) shows its uncertainty. CalRig (reference ± 0.2 °C from 10 to 40 °C) can check the temperature probe in that range but not at module operating temperatures of 60 °C and more, and not the reference cell.

■ 5. Discharge and heat (R11, R13)

Discharge (R11, met). The active path has a 145 ms time constant (174 ms at +20 % capacitance): 100 V falls below 30 V in 0.21 s and below 1 V in 0.80 s, against 2 s. The passive bleed has a 66 s time constant (79 s at +20 %): 100 V falls below 60 V in 40 s, against 60 s. Peak dump power is 455 W and peak dump current 4.5 A for milliseconds; the bleed draws 10 mA (1.0 W) at 100 V, which firmware subtracts from the current reading. At 99 % of Voc the module still delivers up to 2.51 A (rating limit case), more than the 0.5 A at which R11 allows the circuit to open. Firmware must keep the load switch closed until the current falls below 0.5 A, at most 6 ms more with +20 % capacitance. With that rule R11 is met.

Stored energy. 33 J at 100 V (22 J with 4.4 mF). At one sweep every 5 s the dump resistor averages 1.6 W for the reference module and 6.6 W at the 100 V limit.

Heat (R13, at risk). The case has 0.113 m² of outside area and a 0.0286 m² lid, so it sheds 1.02 W/K. In full sun at 45 °C ambient:

Table 4. Inside temperature, lumped model

Case color	Reference module, one sweep per 5 s	100 V limit, one sweep per 5 s
Light grey (absorptance 0.30)	11.0 W, about 56 °C	16.0 W, about 61 °C
Dark (absorptance 0.90)	28.2 W, about 73 °C	33.2 W, about 78 °C

Solar gain on the lid dominates. Even a light case reaches about 56 °C inside, above the usual 45 °C charging limit of a Li-ion cell and close to the limits of commodity display boards. The design therefore needs a light-colored case, a charger with a temperature cut-off (added to bom/bom.csv line 10), and the advice to keep the tracer in shade

between sweeps. A printed sun hood over the display is now fitted (PVT-DDR-002 item 12). It shades the window on three sides and from above; its small shading of the lid is ignored here, so the figures above stay conservative. Display readability in direct sun is still **not met** on paper: a hood improves contrast but readability of the adopted TFT (decision 5) can only be shown in a field check, which is TRL 4 work.

■ 6. Battery (R14)

The controller and display draw 0.60 W and the isolated side 0.25 W at 5 V, or 0.94 W from the cell through the boost. A 3,000 mAh cell with 90 % usable (9.7 Wh) gives **10.3 h**, about 1,853 sweeps at 20 s each. R14 (8 h or 200 sweeps) is met. The TRL 2 figure of 16 h left out the isolation and the boost.

■ 7. Mass, size and cost (R12, R17)

Table 5. Mass estimate

Item	Mass (g)
Enclosure body and lid, ABS 3 mm, with bumpers and glands	402
Load capacitors, 3 x 75 g	225
Test leads, 2 x 1 m 4 mm ² with MC4	162
Sensor pod with 3 m cable	150
DC isolator	120
Battery, holder, charger	72
Hardware and consumables	50
Controller and display	45
Dump and bleed resistors	45
Measurement board; MOSFETs and bar; fuse and holder	40 each
Isolation board	10
Display sun hood, PETG, with screws	17
Window, PC 72 x 54 x 3 mm, and gasket	16
Chassis plate, PC 1.5 mm (model volume 37.8 cm ³)	45
USB-C charging socket with lead	8
Total	1,486

The window was 10 g in v0.2, too low for 3 mm polycarbonate; it is now computed from the model (16 g).

R12 is met: 1.49 kg against 1.5 kg, with about 14 g to spare on estimated masses (accepted by Amish on 2026-10-02, with the prototype weighed at TRL 4; PVT-DEC-001, item 3). Over the bumpers the case is 234 x 144 x 80 mm, 244 mm over the glands and 98 mm to the top of the isolator knob and sun hood, inside 250 x 150 x 100 mm.

The priced BOM totals **USD 178.00**. Value-engineering target: USD 165 (`budget_usd` , a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 178, **USD 13 over the target**. The increase over v0.2 (USD 164) is the parts added to make the design buildable (PVT-DDR-003): the chassis plate (USD 3), the USB-C socket (USD 5), the pod ADC module (USD 3), the pod clip and gland (USD 1) and fixings (USD 2). Savings worth trying are listed in the design decisions register.

■ 8. Results

Table 6. Requirement status (from `docs/04-calcs/results.csv`)

ID	Value	Target	Status
R1	0 to 100 V; high-voltage case 83 V at -10 °C; capacitors 160 V, MOSFETs 150 V	0 to 100 V, below 120 V	Met (design review)
R2	0 to 20.48 A full scale; shunt 1.6 W of 3 W at 20 A	0 to 20 A	Met (design review)
R3	20.5 ms high-current case at 6.6 mF nominal (16.4 ms at -20 %); 32 ms reference; up to 81 ms	20 to 200 ms	Not met (worst-case tolerance of capacitors as bought; a bank selected to 6.4 mF or more at build was decided on 2026-10-02, PVT-DEC-001 item 4, and is not yet calculated)
R4	410 pairs minimum	200 or more	Met
R5	V $\pm 0.17\%$ $\pm 0.024\%$ FS; I $\pm 0.21\%$ $\pm 0.026\%$ FS	$\pm 1\%$ of reading $\pm 0.1\%$ FS	Met on paper (needs calibration)
R6	Instrument noise 0.08 % per sample	Pmax $\pm 1\%$ over 3 sweeps	Not verifiable at TRL 3
R7	Irradiance ± 3.6 to $\pm 5.4\%$; temperature $\pm 2\text{ }^{\circ}\text{C}$ budget	$\pm 5\%$, $\pm 2\text{ }^{\circ}\text{C}$	At risk
R8	± 3.7 to $\pm 5.4\%$ (needs reference cell within $\pm 4.5\%$)	$\pm 5\%$ or better	At risk
R9	About 7 or more samples per volt; rules drafted	Flags within 5 s	Not verifiable at TRL 3
R10	A, B, C, reject thresholds decided (PVT-DDR-002); editable table	Partner-editable grades	Met (design review)
R11	30 V in 0.21 s; 60 V passive in 40 s; opens below 0.5 A after up to 6 ms more	30 V in 2 s; 60 V in 60 s; open below 0.5 A	Met
R12	1.49 kg; 234 x 144 x 80 mm over bumpers	1.5 kg; 250 x 150 x 100 mm	Met
R13	Inside about 56 °C (light case) to 73 °C (dark) at 45 °C in sun; TFT with sun hood, readability unproven; IP54 case	0 to 45 °C, sun, IP54, readable	Not met (display); at risk (heat)
R14	10.3 h, about 1,853 sweeps	8 h or 200 sweeps	Met
R15	CSV fields defined in PVT-PRC-001; no firmware at TRL 3	CSV per sweep, USB or Wi-Fi export	Met (design review)
R16	No insulation test in this instrument	Detect insulation faults	Not met (out of scope)
R17	USD 178 in parts	Value-engineering target USD 165	Over the target by USD 13

Safety: These calculations cover a live DC instrument at up to 100 V and 20 A that stores up to 33 J. The discharge times assume both the active and passive paths are fitted and working; never rely on one alone, and treat the capacitors as charged until the display reads below 30 V. The heat results show the Li-ion cell can exceed its charging limit in sun; charge only in shade with the isolator open.